

# Exercises on Partial Derivatives and Optimization

**Exercise 1.** Find and classify all critical points of

$$f(x, y) = x^2 + 2y^2 - 4x + 8y + 5.$$

**Exercise 2.** Find the gradient vector of

$$f(x, y) = x^3y + 2xy^2$$

at the point

$$(1, -2).$$

**Exercise 3.** Find the tangent plane to the surface

$$z = x^2 - xy + 3y^2$$

at the point

$$(1, 1, 3).$$

**Exercise 4.** Use the chain rule to compute

$$\frac{dz}{dt}$$

if

$$z = x^2 + y^2 + xy,$$

where

$$x = t^2, \quad y = e^t.$$

**Exercise 5.** Find all critical points and determine whether they are relative maxima, minima, or saddle points for

$$f(x, y) = x^3 - 3x + y^2 - 4y.$$

**Exercise 6.** Find all second-order partial derivatives of

$$f(x, y) = x^2e^{xy} + y^3.$$

**Exercise 7.** Find and classify the critical points of

$$f(x, y) = x^4 + y^4 - 4x^2 - 2y^2.$$

**Exercise 8.** A warehouse is to be located at a point minimizing the function

$$f(x, y) = (x - 1)^2 + (y - 2)^2 + (x - 5)^2 + (y - 6)^2.$$

Find the optimal location.